

The effect of temperature anomaly and macroeconomic fundamentals on agricultural commodity futures returns

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ABSTRACT

This paper is the first to examine the dynamic impact of temperature anomaly and macroeconomic fundamentals on agricultural commodity futures returns. Using a quantile regression approach, we report that temperature anomaly exerts a significant negative (positive) impact on returns of soybean, corn, cotton, and coffee (soybean, corn, and cocoa) futures in extreme bearish and bullish market conditions. Moreover, agricultural commodity futures returns appear to positively respond to aggregate stock returns and negatively influenced by exchange rate changes. However, such results vary across quantiles. We also report that economic activities and various uncertainty measures do not have a strong impact on the returns of agricultural commodity futures. Our study suggests that an adjustment in risk evaluation encompassing weather shocks would benefit portfolio management.

1. Introduction

The increased financialization of agricultural commodities has made their prices sensitive to price movements of major asset classes and other commodities. Agricultural commodity prices are found to have a relationship with prices in aggregate equity (Andreasson et al., 2016), energy (Cabrera and Schulz, 2016), and precious metal (Kang et al., 2017) markets. Furthermore, agricultural commodity markets are increasingly becoming more responsive to macroeconomic variables, such as the U.S. dollar (USD) exchange rates (Nazlioglu and Soytaş, 2012) and macroeconomic factors (Joëts et al., 2017), particularly after the food-price crisis of 2006 and the global financial crisis (GFC) of 2007–2008. Instead of only being considered consumption commodities, agricultural commodities have been extensively traded on futures exchanges globally. Therefore, examining the return dynamics of agricultural commodity futures has implications for farmers, consumers, traders and investors seeking superior risk-adjusted returns and hedging possibilities. This paper aims to examine the effect of temperature anomaly and macroeconomic fundamentals on agricultural commodity futures

returns under different market states.

This paper is primarily motivated by the fact that the world has witnessed meteorological extremes over the last decades (Field et al., 2012). For instance, the 2000s was the warmest decade since 1850, and we also experienced climatic anomalies, such as the European heatwave in 2013 and droughts in Russia and the United States in 2010 and 2012, respectively (Chatzopoulos et al., 2019). These extreme weather events are often linked to high volatility in agricultural commodity prices (van Oordt et al., 2021). For example, the corn and soybean futures prices remain relatively stable for the first few months of 2012. However, due to exceptionally dry months of June and July resulting in a prolonged drought in the United States, prices increased rapidly (see Appendix Fig. A1).

One of the main roles of the agricultural commodity futures market is to provide a hedging or managing price risk tool. This phenomenon limits the risk of potential losses caused by adverse price movements. For example, a wheat farmer is exposed to the risk of financial loss if the wheat price falls before harvest. The farmer can hedge this risk by selling wheat futures that guarantee a predetermined price. However, futures

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prices may exhibit unfavorable movements arising from many factors, including temperature anomalies. Therefore, abnormal meteorological conditions imply greater market uncertainty.¹ The Intergovernmental Panel on Climate Change (IPCC) has stressed that future occurrences of extreme weather events are highly likely. These events are then expected to affect the global food system and the agricultural commodity market. Thus, examining the interaction between temperature anomaly and the agricultural commodity futures market is worthwhile.

The relevant literature has several strands. The first strand, a relatively large one, explores the sensitiveness of agricultural commodity futures to other asset markets (Andreasson et al., 2016; Cabrera and Schulz, 2016; Kang et al., 2017), macroeconomic risk (Joëts et al., 2017), macro news (Caporale et al., 2017), and financial speculation (Andreasson et al., 2016; Sanders et al., 2010), among others. These studies, however, primarily concentrate on a mean-to-mean relationship, ignoring the potential changing pattern in the relationship when the agricultural commodity market switches between different conditions. The second one focuses on the economic effect of weather shocks and weather risk management. For instance, researchers show the impact of weather shocks on energy futures (Mu, 2007), weather shocks' business cycle implications (Gallic and Vermandel, 2020), the macroeconomic transmission of El Niño weather shocks (Cashin et al., 2017), and extreme weather event's potential impact on European cities (Jahn, 2015), among others. Further, empirical studies report that weather risk management through weather derivatives increases the value of weather-sensitive firms (Pérez-González and Yun, 2013), and investing in a decarbonized index provides investors a "free option on carbon" (Andersson et al., 2016). The third strand explores uncertainty, risk, and return dynamics in the commodity (agricultural commodities in particular) futures markets (see, for example, Gorton et al., 2013; Carbonez et al., 2009; Dincerler et al., 2020; Moskowitz et al., 2012).

Despite a reasonably large literature as mentioned in the previous paragraph, to the best of our knowledge, there is no previous study exploring the effect of temperature anomaly on agricultural commodity futures returns, which is the main focus of the present paper. We model the conditional return distribution of eight agricultural commodity futures (cotton, corn, rapeseed, soybean, coffee, cocoa, wheat, and sunflower) to explore how they are shaped by temperature anomaly (global temperature deviations from the average of a reference period), other asset returns (aggregate stock returns and exchange rate changes), macroeconomic factor (economic activity), and uncertainty measures (economic policy uncertainty, equity market implied volatility, and oil price volatility). To explore the dynamic relationship of agricultural commodity futures returns with temperature anomaly and macroeconomic fundamentals, we adopt a quantile regression (QR henceforth) modeling approach, following Koenker and Bassett Jr (1978). This approach enables us to examine the co-movement (Baur, 2013; Mensi et al., 2014) and causal relationship (Reboredo and Uddin, 2016) between agricultural commodity futures prices and the explanatory variables. Moreover, it allows us to investigate the relationship in different conditions within the agricultural commodity market, such as bearish (i.e., low quantile), normal (i.e., median quantile), and bullish (i.e., high quantile).

We extend the existing literature in several aspects. First, our paper is the first to provide evidence on the effect of temperature anomaly on agricultural commodity futures prices. This evidence may lead to incorporating climate change factors into portfolio choices. If temperature anomaly is found to be a risk factor (if it is a factor that drives

agricultural commodity futures returns), investors may construct a well-diversified portfolio that isolates the exposure to that risk factor. As such, the climate risk can be hedged by trading this portfolio (Engle et al., 2020).

Second, we examine both the contemporaneous and dynamic relationship by exploring how this relationship varies in different market conditions. This analysis provides important insights in predicting agricultural commodity prices and deriving risk management and hedging strategies for commodity investors. Understanding the dynamic relationship is important to develop a dynamic hedging approach as temperature anomaly, which is part of the overall climate risk, consists of a long-term phenomenon. Further, an asymmetric relationship in different market conditions potentially help investors changing asset allocation and portfolio composition as the market switches from one state to another. This is based on investors' assessment of risk level and corresponding expected returns in different quantiles.

Third, we also consider a large set of explanatory variables to provide robust evidence of the relationship between temperature anomaly and agricultural commodity futures returns. The choice of the explanatory variables used is dictated by economic rationale and empirical results found in the relevant existing literature (this point is discussed in more detail in Section 4). As indicated previously, we control for the overall equity returns, exchange rate changes, business, economic policy, equity market, and oil price uncertainty. Our approach also accounts for the impact caused by the food-price crisis of 2006.

Our empirical findings indicate robust evidence of the determinants of agricultural commodity futures prices. First, we find an asymmetric impact of the temperature anomaly across the return quantiles of agricultural commodity futures. For instance, the temperature anomaly has a significant negative effect on the returns of soybean, corn, and cotton futures in low quantiles (0.05–0.10) as well as coffee in high quantiles (0.90–0.95), and a positive impact on the returns of soybean, corn, wheat, cocoa, and cotton futures in high quantiles (0.90–0.95). Second, in the cases of corn, rapeseed, wheat, cotton, and coffee, we report a significant positive relationship between agricultural commodity futures returns and aggregate stock returns in low quantiles, typically between 0.05 and 0.10.

Third, agricultural commodity returns are found to respond negatively to exchange rate changes. This result is statistically significant in all quantiles for rapeseed and cotton, in the extreme low quantile (0.05) for coffee, from 0.25 to 0.75 quantiles for soybean, and from 0.25 to 0.95 quantiles for sunflower futures. The second and third findings indicate potential diversification opportunities for agricultural commodity investors once equities and currencies are incorporated into the portfolio. Nonetheless, such an opportunity is sensitive to particular conditions of the agricultural commodity futures market. Fourth, changes in economic activities and uncertainties associated with economic policy, equity market, and oil prices typically do not have a strong impact on agricultural commodity futures returns.

Following this introduction, the paper is divided into five sections. Section 2 presents a brief review of the relevant literature. Section 3 details the methodology adopted. Section 4 presents the data and preliminary results. Section 5 reports and discusses empirical results. Section 6 contains the concluding remarks, limitations of this study and suggestions for future research.

2. Literature review

The relevant empirical studies may be divided into three groups. The first strand explores the interaction between agricultural commodity futures, other asset classes, and macroeconomic factors. The second group focuses on weather shocks, weather derivatives, and weather risk management. The third one consists of the studies on uncertainty and risk in the commodity futures market.

¹ In the present paper, the concepts of uncertainty and risk are distinguished and used according to the definitions found in the respective relevant literature. The outcome in both cases (i.e. uncertainty and risk) are unknown prior to the event, although the probability distribution behind all possible outcomes is known in the case of risk but not known in the case of uncertainty. For a detailed discussion, see Rachev et al. (2011).

2.1. Interaction of agricultural commodity futures with other asset classes, and macroeconomic factors

The correlation and interdependence among agricultural commodity futures prices are examined by several studies (Malliaris and Urrutia, 1996; He and Chen, 2011; Ke et al., 2019). These papers, in general, show evidence of long-term relationship/ cross-correlation across commodities and markets. For example, Malliaris and Urrutia (1996) report the absence of short-term causality but find a strong long-term relationship among six agricultural commodities. He and Chen (2011) report significant cross-correlation between the Chinese and U.S. agricultural futures markets, reporting that such markets are found to share similar characteristics under several conditions. Similar results are found in Ke et al. (2019), showing a significant upside risk transmission between the Chinese and U.S. commodity markets.

A relatively large literature explores the interaction of agricultural commodity futures with other asset classes such as equity (Andreasson et al., 2016), energy (Han et al., 2020; Tiwari et al., 2020; Kang et al., 2019), precious metal (Kang et al., 2017), and exchange rates (Nazlioglu and Soytas, 2012). Andreasson et al. (2016) report that agricultural commodities (e.g., corn, wheat) Granger-cause aggregate equity returns; however, the causality is reversed in the case of soybean. Further, agricultural commodity prices are not found to be highly sensitive to equity market volatility.

This strand of literature is dominated by studies focusing on the relationship between agricultural commodity futures and energy prices. Cabrera and Schulz (2016) show that agricultural commodity and energy prices move in tandem in the long run. Similar results are also reported in Balcombe and Rapsomanikis (2008) and Hassouneh et al. (2012), among others. Du et al. (2011) report volatility spillover between crude oil and agricultural commodities, such as corn and wheat. Han et al. (2020) also find significant volatility linkages between agricultural futures and energy returns. Nevertheless, Fowowe (2016) argues that agricultural commodity prices are invariant to oil price changes in South Africa. Kang et al.'s (2019) findings indicate a bidirectional and unsymmetrical interdependence between agricultural commodities and oil prices at distinct frequency ranges. Tiwari et al. (2020) report material relationships between fuel, food, industrial, and metal prices. Their volatility spillover analysis suggests that the agriculture market is particularly impacted due to disturbances experienced by other markets. The interdependence between agricultural commodities and energy prices is also examined in Ji et al. (2018), Yahya et al. (2019), Luo and Ji (2018), among others.

The information transmission between precious metal and agricultural commodity prices is explored in Kang et al. (2017), finding that precious metals (i.e., gold and silver) are mostly information transmitters, while agricultural commodities (i.e., corn, wheat, and rice) are mostly receivers of spillover. Nazlioglu and Soytas (2012) find that the weak U.S. Dollar exerts a positive impact on agricultural commodity prices. Liu et al. (2020) show that average commodity (including agricultural commodities) returns have predictive power on exchange rate returns in Australia, Canada, New Zealand and South Africa.

Several other studies explore the effect of macroeconomic factors and macro news over agricultural commodity price fluctuations. Joëts et al. (2017) find that the agricultural commodity markets are susceptible to changes in macroeconomic factors. Caporale et al. (2017) explore the linkages between macro news and commodity returns, including several agricultural commodities in their sample (e.g., corn, wheat, and soybean). They find mean and variance spillover from macro news to commodity returns. Variance spillover, however, is more significant, and these linkages are particularly evident after the GFC of 2007–2008. Li et al. (2017) report that price bubbles in the Chinese agricultural commodity market positively respond to economic growth, money supply, and inflation rate while reacting negatively to changes in the interest rate. Nam (2021) shows that agricultural commodities experience inflationary pressure resulting from climate uncertainty.

They further show that individual commodities (such as maize and soybean), compared to aggregated commodity indices, are more sensitive to the effect of climate uncertainty.

2.2. Weather shocks, weather derivatives, and weather risk management

This strand of literature focuses on the economic effect of weather shocks (e.g., high temperature, prolonged droughts) and weather risk management approaches. Mu (2007) finds that weather shocks influence the conditional mean and volatility of returns of the U.S. natural gas futures. Dell et al. (2012) show that temperature shocks reduce economic growth, agricultural and industrial output, and political stability, particularly in developing countries. However, Burke et al. (2015) contrast such a finding, reporting that the relationship between economic productivity and temperature is non-linear in both wealthy and developing countries. Donadelli et al. (2017) show that temperature shock exhibits a statistically significant negative impact over total factor productivity, output, and labor productivity. Fomby et al. (2013) find that while natural disasters' (e.g., droughts, floods, earthquakes, and storms) effect on developing countries is stronger than on developed countries, some natural disasters exert even positive effects on economic growth. A similar result is found in Cashin et al. (2017) regarding the impact of El Niño. The authors find that some countries (such as Australia, New Zealand, and South Africa) experienced a short-term reduction in economic activity while others experienced a growth-enhancing effect of El Niño. Gallic and Vermandel (2020) report that weather shocks (i) increase macroeconomic volatility, and (ii) it can explain 35% of GDP and agricultural output fluctuations. The economic impact of extreme weather events is also reported in Jahn (2015).

With the significant evidence of weather shocks' effect on aggregated and disaggregated economy, researchers and practitioners propose and develop several approaches to managing weather risk. For instance, weather derivatives were first introduced in 1997 in the United States to hedge weather risk (Brockett et al., 2005). Pérez-González and Yun (2013) show that weather risk management through weather derivatives increases the value of weather-sensitive firms. The role of weather derivatives in weather risk management (Brockett et al., 2005), valuation and pricing of weather derivatives (Alaton et al., 2002; Benth and Benth, 2007), and the market price of weather risk (Härdle and Cabrera, 2012) are also explored in the literature. Andersson et al. (2016) present a dynamic investment strategy of holding a low-carbon portfolio that performs as good as the benchmark. In addition, De Jong and Nguyen (2016) show that bond investors may use traditional portfolio construction techniques to hedge against climate uncertainty. These authors further argue that such an approach does not expose to the risk of offsetting benchmark-tracking properties of the portfolio. Engle et al. (2020) propose a model to hedge climate risk dynamically. The authors show that their approach results in parsimonious and industry-balanced portfolios that perform well in hedging climate change news.

2.3. Uncertainty and risk in the commodity futures market

Researchers have been increasingly interested in exploring uncertainty, risk, and return dynamics in the commodity futures markets on the back of the financialization process. Gorton et al. (2013) report that risk premiums in the commodity futures market are time-varying and depend on the level of physical commodities. Similar result is found in Carbonez et al. (2009) and Dincerler et al. (2020), among others. Moskowitz et al. (2012) decompose commodity futures return into spot return and roll return. Roll return is earned by maintaining a constant exposure to a particular commodity futures market. Gorton and Rouwenhorst (2006) and Gorton et al. (2013), among others, report that investment in commodity futures generates "equity-like" returns and provides portfolio diversification benefits.

It is often argued that the agricultural futures markets are

predominantly hedging markets, where speculation plays an important role in driving agricultural commodity futures volume, prices, and returns (Working, 1961). Andreasson et al. (2016) report evidence of linear causality running from agricultural commodity prices to excess speculation. However, Sanders et al. (2010) do not find evidence that speculation drives agricultural commodity prices. The adverse effect of speculation on the agricultural commodity futures market is also rejected in Dimpfl et al. (2017). More recently, Palazzi et al. (2020) explore whether speculative trading influences the price of commodities or if speculators are trend traders even under distinct market conditions (e.g., bullish, bearish). Their empirical findings point to the absence of causality in such hypotheses.

Reviewing the literature, we observe two main gaps. First, it essentially ignores the effect of either weather or temperature anomaly on agricultural commodity futures returns. Second, examining the dynamic relationship of agricultural commodity prices with macroeconomic variables and risk measures across quantiles was beyond the scope of previous studies. Thus, the present study aims to fulfill these gaps.

3. Method

We observe that the relevant researchers have adopted several empirical methods. Before explaining the methodology adopted in this paper (QR approach), we summarize the methods used by our predecessors and justify why the QR approach optimally fits our research context. The existing literature extensively uses vector autoregressive or multivariate generalized autoregressive conditional heteroskedasticity based models to explore interaction, interdependence, or causality across markets (see, for example, Andreasson et al., 2016; Bakas and Triantafyllou, 2018; Caporale et al., 2017; Cashin et al., 2017). However, these approaches cannot explore quantile dependence or relationship between financial markets as they are reliant on conditional variance modeling (Jiang et al., 2016). Since cross-market correlation can manifest interdependence, researchers also use dynamic conditional correlation (DCC) based models (see, for example, Cabrera and Schulz, 2016; Creti et al., 2013). However, the DCC-based approach can be inadequate due to its restrictive assumption of smoothing parameters (Labidi et al., 2018). Several papers have also used a copula-based approach to examine the dependence between agricultural commodity and other asset prices (see, for example, Yahya et al., 2019; Ji et al., 2018). Copulas, however, in general measure a time-statistic dependence structure (Labidi et al., 2018) but they may suffer from a lack of parameter choices (Embrechts et al., 2001).

In the present study, we apply the conditional QR approach proposed in Koenker and Bassett Jr (1978) to explore the relationship among agricultural commodity futures returns with temperature anomaly and macroeconomic fundamentals. Several factors dictate the use of the QR approach. First, this approach allows us to examine if the conditional mean of an independent variable explains any conditional quantile (i.e., not only conditional mean) of the dependent variable. Therefore, as opposed to the ordinary least squares (OLS) method that examines a mean-to-mean relationship, the QR approach captures the entire distribution of the dependent variable by splitting the dependent variable into quantiles (τ).

Second, this approach is particularly useful in our study, as our preliminary analysis reveals several spikes in the agricultural commodity futures prices, especially after the advent of the food-price crisis of 2006 and in response to extreme weather events.² These spikes indicate that agricultural commodity prices have experienced large fluctuations that may have affected their relationship with the explanatory variables. In addition, the QR approach is less sensitive to outliers

compared to the traditional OLS approach (Lien et al., 2016).

Third, through the QR method, it is possible to observe the structure of dependence and co-movement (Baur, 2013; Mensi et al., 2014) as well as Granger causality (Chuang et al., 2009) between two variables under different market conditions (Reboredo and Uddin, 2016). The coefficients in low τ (i.e., $\tau \in [0.10]$), median τ (i.e., $\tau \in [0.50]$), and high τ (i.e., $\tau \in [0.90]$) reveal the effect that explanatory variables have on the dependent variable in bearish, normal, and bullish market conditions, respectively. By isolating such effects in both extreme and normal market conditions, the QR approach provides more information in terms of the relationship between a dependent and an independent variable.

In our study, we explore how a set of explanatory variables influence the conditional τ of the agricultural commodity futures return distribution y_t , namely $Q_{y_t}(\tau|y_{t-1}, X_t, X_{t-1}, d_t)$. Hence,

$$Q_{y_t}(\tau|y_{t-1}, X_t, X_{t-1}, d_t) = \alpha_\tau + \varphi_\tau y_{t-1} + \beta_\tau X_t + \eta_\tau X_{t-1} + \phi d_t + u_{\tau t} \quad (1)$$

where y_t consists of the agricultural commodity futures returns, y_{t-1} is the lagged value of the dependent variable, X_t and X_{t-1} are the $(k \times 1)$ vector of temperature anomaly and macroeconomic fundamentals at time t and $t - 1$, respectively, and $u_{\tau t}$ denotes the residual vector. The dummy variable that incorporates the effect of the food-price crisis of 2006 is represented by d_t , which takes a value of zero until December 2005 and one afterwards. This dummy variable, therefore, captures the permanent effect of the food-price crisis on the conditional quantiles of agricultural commodity futures returns.

The parameters α_τ , φ_τ , and ϕ capture the unconditional quantile. They respectively manifest the effect of the lagged agricultural commodity futures returns and the food-price crisis of 2006 on the return quantiles. The parameter β_τ for $\tau \in [0, 1]$ is a parameter vector that represents the effect of explanatory variables on the dependent variable. If β_τ is statistically different from zero for a particular τ , it indicates that the explanatory variable has a statistically significant impact on the dependent variable in that particular τ . Furthermore, if β_τ remains the same across the τ , the relationship would be constant across different market conditions.

On the other hand, if β_τ values increase (decrease) across τ , the relationship is considered a progressively stronger (weaker) one. In the case that β_τ values are the same (different) across τ , the relationship is considered to be symmetric (asymmetric) across different market conditions (Baur, 2013; Mensi et al., 2014). Lastly, the η_τ for $\tau \in [0, 1]$ is a parameter vector that explores Granger causality between the dependent and explanatory variables. If η_τ values are not statistically different from zero, it indicates that past information embedded in the explanatory variables does not explain conditional quantiles of the dependent variable. Therefore, there is no evidence of Granger causality. The opposite is valid when η_τ values are statistically different from zero.

In this study, we use seven conditional quantiles $\tau = \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$ to examine the impact of the explanatory variables on the dependent variable under different market conditions. The motivation behind including a lag of each of the independent variables is to assess the dynamic relationship between agricultural commodity futures returns and the explanatory variables. We also consider longer lags, although the impact of the explanatory variables on agricultural commodity futures returns are found not to be statistically significant.³ For a given quantile τ , the regression model in Eq. (1) is estimated using a function that minimizes the weighted absolute deviation, as follows:

$$\arg \min_{\alpha(\tau), \varphi(\tau), \beta(\tau), \eta(\tau), \phi(\tau)} \sum_{t=1}^T \rho_\tau(y_t - \alpha_\tau - \varphi_\tau y_{t-1} - \beta_\tau X_t - \eta_\tau X_{t-1} - \phi_\tau d_t) \quad (2)$$

$$\rho_\tau(u) = u[\tau - I(u < 0)] \quad (3)$$

² The time-trends of the agricultural commodity futures prices are presented in Appendix Fig. A2. The descriptive statistics of the variables used in this paper are displayed in Table 3.

³ These results are not reported in the present paper to conserve space.

where $0 < \tau < 1$ and $I(\cdot)$ is the check function of the residual vector. The minimization problem in Eq. (2) is solved by linear programming methods.⁴ Standard errors for the parameters are estimated using the design matrix bootstrapping procedure suggested in Buchinsky (1995). The procedure draws samples from the pairs of regressors and estimates a design matrix from which it calculates standard errors. This method is beneficial for our sample, which is relatively small, and some of the return series do not follow a Gaussian distribution (Mensi et al., 2014).

4. Data and preliminary analysis

We use monthly data covering a period of 20 years, spanning from January 1999 to December 2018. This period is chosen due to the presence of relevant or even disruptive events that led to high volatility in commodity prices, such as the 2000s energy crisis, food-price crisis of 2006, and the GFC of 2007–2008. The dependent variables consist of the returns of agricultural commodity futures with continuous settlement prices. Continuous settlement prices are adopted because multiple contracts are traded on the same commodity with different maturities. We consider eight agricultural commodities: coffee, cotton, corn, rapeseed, soybean, cocoa, wheat, and sunflower.⁵ These commodities are the most traded agricultural commodities. They are also commonly explored in the relevant literature (e.g., Natanelov et al., 2011; Nazlioglu and Soytaş, 2012; Hasanov et al., 2016). The details of the agricultural commodity futures are reported in Table 1, showing their classification, the exchange in which they are traded, the number of units per contract, and respective DataStream ticker.

Due to possible linkages between temperature anomaly and agricultural commodity futures prices, we consider two widely used datasets containing temperature anomaly time series. They are (i) the Climatic Research Unit of the University of East Anglia and Hedley Center of the UK Met Office anomaly data (HadCRUT4) and (ii) the NASA Goddard Institute for Space Studies Land-Ocean Temperature Index (LOTI). These data are essentially measures of the global temperature deviations from the average of a reference period. LOTI and HadCRUT4 measure deviations from 1951 to 1980 and from 1961 to 1990 averages, respec-

Table 1
Commodity futures contracts.

Commodity	Class	Exchange	Units/ contract	DataStream code
Coffee	Softs	Intercontinental Exchange	Pounds/ 37,500	NKCCS00
Cocoa	Softs	Intercontinental Exchange	Metric ton/10	NCCCS00
Cotton	Softs	Intercontinental Exchange	Pounds/ 50,000	NCTCS00
Corn	Grains & Oilseeds	Chicago Board of Trade	Bushels/ 5000	CCFCS00
Rapeseed	Grains & Oilseeds	Marché à Terme International de France	Metric ton/50	PROCS00
Soybean	Grains & Oilseeds	Chicago Board of Trade	Bushels/ 5000	CSYCS00
Sunflower	Grains & Oilseeds	South African Futures Exchange	Metric ton/50	SAUCS00
Wheat	Grains & Oilseeds	Kansas City Board of Trade	Bushels/ 5000	KKKCS00

⁴ We use the quantreg package in R with 1000 iterations.

⁵ Initially, we also considered rough rice and oats futures. However, they were excluded from the sample as futures price data underlying these two commodities were only available for a short sample period. In the cases of coffee, wheat, corn, and soybean futures, we considered price series from different exchanges. However, results were found to be qualitatively similar.

tively. These proxies for temperature anomaly are widely used by researchers. For example, HadCRUT4 has been used in Morice et al. (2012), Hansen et al. (2010), and Tung and Zhou (2013). Further, Schmidt et al. (2014) and Miller et al. (2021), among others, have used LOTI. Since we obtain qualitatively similar results for both temperature anomaly data, our empirical results are based on the HadCRUT4 dataset. The empirical results pertaining to LOTI may be obtained from the corresponding author upon request.

We consider six macroeconomic determinants of agricultural commodity futures prices. Firstly, we include the broad U.S. stock market index (S&P500), as it is widely used in the literature as a determinant of commodity futures prices. Secondly, a trade-weighted USD against an index composed of major currencies is considered. This index measures the relative strength of the USD compared to the exchange rates of currencies in the Euro area, Canada, Japan, the UK, and Switzerland. Thirdly, the Auroba-Diebold-Scotti Business Conditions Index (ADS) is used as a proxy for economic activity. This index is derived from several economic indicators, such as weekly initial jobless claims, monthly payroll employment, industrial production, personal income less transfer payments, manufacturing and trade sales, and quarterly real gross domestic product (GDP).

Fourthly, we include three uncertainty measures as potential drivers of agricultural commodity futures prices. The first one is the economic policy uncertainty (EPU) index, which is estimated based on the frequency of publication of a group of words related to economic policy uncertainty in leading newspapers. This index captures uncertainties related to whether a policy decision is made and its subsequent economic impact. The second uncertainty measure is the oil price implied volatility (OILV) derived from the WTI crude oil spot prices. Due to the lack of observations for the energy sector volatility, we estimate a univariate GARCH(1,1) model on crude oil returns following Bollerslev (1986), adopting the squared residuals as a proxy for the conditional volatility of the energy market. The third uncertainty measure is the equity market implied volatility index (VIX), which reflects the expected volatility of the S&P500 stock prices. More specifically, this index captures market expectations regarding the fluctuation of the S&P500 in the subsequent 30 days. All sources of data are mentioned in Table 2.

The choice of the explanatory variables used in the present study is dictated by economic rationale and empirical results in the relevant literature. Firstly, temperature anomaly or climate change may impact food production, population growth, change in dietary behavior,

Table 2
Data sources.

Variables	Data sources
Agricultural commodity futures prices	DataStream
Temperature anomaly LOTI HadCRUT4	NASA Goddard Institute for Space Studies Climatic Research Unit, University of East Anglia
Macroeconomic variables S&P 500 USD ADS EPU VIX OILV	DataStream DataStream Federal Reserve Bank of Philadelphia Economic policy uncertainty webpage Chicago Board of Exchange Authors' own estimation

Notes. LOTI and HadCRUT4 are variables representing temperature anomaly. These variables are essentially measures of the global temperature deviations from the average of a reference period. LOTI and HadCRUT4 respectively measure deviations from 1951 to 1980 and 1961 to 1990 temperature averages. S&P 500: the broad U.S. stock market index; USD: a trade-weighted U.S. Dollar to major currencies index; ADS: the Auroba-Diebold-Scotti Business Conditions Index; EPU: the Economic Policy Uncertainty Index; VIX: the Equity Market Implied Volatility Index; OILV: Oil Price Implied Volatility.

environmental degradation, and trade relationship (Janetos et al., 2017). Such phenomena are likely to affect the global food system and agricultural commodity prices. Chatzopoulos et al. (2019) provide evidence of the asymmetric response of crop prices to positive and negative agrometeorological shocks.

Secondly, in recent years, there has been a substantial increase in investments in the commodity markets. This investment growth has led the commodity markets to an increasingly larger vulnerability to market-wide shocks, such as changes in equity prices and exchange rates (Tang and Xiong, 2012). Nazlioglu and Soytaş (2012) and Andreasson et al. (2016), among others, show that agricultural commodity prices respond positively to weak USD and aggregate equity returns.

Thirdly, an expansion in economic activity may increase the demand for agricultural commodities, which ultimately should push up commodity prices (Frankel and Rose, 2010). Several papers report the relevant role of crude oil price volatilities in determining agricultural commodity prices (e.g., Balcombe and Rapsomanikis, 2008; Hassouneh et al., 2012; Nazlioglu and Soytaş, 2012; Cabrera and Schulz, 2016). This linkage may arise since agricultural commodity prices vary in response to change in energy prices, once such changes impact the production, processing, and transportation costs of agricultural commodities. Fourthly, uncertainty measures may impact returns of agricultural commodity futures following changes in the degree of risk aversion of investors and shift of investment across asset classes. For instance, Bakas and Triantafyllou (2018) show that macroeconomic and financial uncertainty impact commodity prices.

Table 3 reports descriptive statistics of the variables. We observe that among the agricultural commodity futures, sunflower (0.29%) and coffee (−0.03%) contracts generate the highest and lowest average returns, respectively. Coffee futures returns also exhibit the highest volatility. Overall, volatilities for the agricultural commodity futures returns range between 6.4% (rapeseed) and 9.46% (coffee). Among the eight agricultural commodity futures, four (cotton, corn, rapeseed, and soybean) show negative skewness, indicating the presence of extreme negative returns without corresponding extreme positive returns.

The remaining four commodity futures (coffee, cocoa, wheat, and sunflower) show positive skewness. The kurtosis statistics are greater than three in all cases, indicating leptokurtic distributions. The means of the temperature anomaly variables are positive, indicating an overall rise in the global temperature. Regarding macroeconomic variables, the S&P500, EPU, and OILV show positive mean changes, while the USD,

ADS, and VIX manifest negative mean changes. The null hypothesis of normality is rejected for most of the variables, except for cocoa, sunflower, and USD. All the variables consist of stationary stochastic processes, as the null hypothesis of a unit root is rejected by the stationarity tests (Dickey and Fuller, 1979; Phillips and Perron, 1988) at the 1% significance level.

5. Results and discussion

This section summarizes the interaction of agricultural commodity futures returns with temperature anomaly and macroeconomic fundamentals. Although our main results are derived from estimating the model detailed in Eq. (1) using the QR approach, we first present our baseline results derived from the ordinary least squares (OLS) regression approach.

5.1. OLS regression results

The OLS regression results are presented in Table 4. We observe that the lagged dependent variable has a significant negative impact on the current returns of cocoa and coffee futures. This result indicates that the past returns may be used to predict their respective future returns for these two commodity futures. However, information embedded in the past returns is diffused immediately in the present returns for the remaining commodity futures. Therefore, past returns unlikely guide forecasting future returns.

Among other variables, the S&P500 and USD appear to be significant determinants of agricultural commodity futures returns. We find that soybean, sunflower, corn, wheat, and coffee futures returns respond positively to the S&P500 stock returns, while soybean, sunflower, rapeseed, and cotton futures returns respond negatively to the USD. These results imply that agricultural commodity futures may not be an appropriate hedge for equity investors. However, currencies may be included in agricultural commodity portfolios for diversification purposes. Changes in the economic activity (ADS) only have a marginally significant impact on the sunflower and rapeseed futures returns. Temperature anomaly and uncertainty measures appear not to have any statistically significant influence on the dependent variable, except for a negative impact of OILV on corn futures returns.

Regarding the effect of one-period lagged explanatory variables, the USD's impact on sunflower and rapeseed futures returns persists until

Table 3
Descriptive statistics.

	Mean	Standard deviation	Skewness	Kurtosis	J-B	PP	ADF
Panel A: Agricultural commodity futures							
Coffee	−0.0003	0.0946	0.4848	3.9451	18.2592***	−18.7137***	−11.0990***
Cocoa	0.0020	0.0914	0.0763	3.5068	2.7899	−19.0075***	−10.3140***
Cotton	0.0012	0.0876	−0.1185	4.0680	11.9190***	−16.1115***	−4.5724***
Wheat	0.0020	0.0855	0.1868	3.9619	10.6044***	−16.7818***	−16.7782***
Corn	0.0024	0.0825	−0.4326	3.7529	13.1018***	−15.2462***	−7.7674***
Rapeseed	0.0018	0.0640	−0.3827	4.2019	20.2176***	−14.7496***	−6.1771***
Soybean	0.0022	0.0820	−0.6076	4.4742	36.3466***	−15.8562***	−6.9100***
Sunflower	0.0029	0.0786	0.1062	3.0536	0.4759	−13.3921***	−13.4764***
Panel B: Temperature anomaly							
LOTI	0.0026	0.2106	0.0621	4.9861	39.4353***	−36.7960***	−14.0680***
HadCRUT4	0.0023	0.2374	−0.0158	5.6551	70.2124***	−34.2019***	−6.9161***
Panel C: Macroeconomic fundamentals							
S&P 500	0.0034	0.0459	−0.8052	5.1367	71.2929***	−15.0585***	−8.4483***
USD	−0.0001	0.0167	0.0392	3.5922	3.5536	−10.7280***	−6.1125***
ADS	−0.0003	0.2906	−0.2232	5.4815	63.3061***	−12.0139***	−5.9915***
EPU	0.0022	0.1806	0.7715	5.8573	105.0086***	−26.6685***	−11.3864***
VIX	−0.0017	0.1990	0.6871	5.0678	61.3845***	−22.5451***	−8.7947***
OILV	0.0061	0.0987	−0.5494	4.2974	28.6645***	−13.1426***	−13.2119***

Notes. J-B is the test statistic for the null hypothesis of normality. PP and ADF are respectively Phillip-Perron (Phillips and Perron, 1988) and augmented Dickey-Fuller (Dickey and Fuller, 1979) test statistics for the null hypothesis of a unit root. See notes to Table 2 for the definitions of the variables. *** indicates statistical significance at the 1% level.

Table 4

Agricultural commodity futures, temperature anomaly, and macroeconomic fundamentals: OLS regression.

	Soybean	Sunflower	Rapeseed	Corn	Wheat	Cocoa	Cotton	Coffee
Lagged dependent variable	-0.035	0.006	-0.001	0.037	-0.097	-0.272***	-0.111	-0.244***
S&P 500	0.270*	0.260**	0.160	0.402**	0.335**	0.190	0.307	0.419**
USD	-0.912***	-1.015***	-0.932***	-0.547	-0.352	-0.421	-1.105***	-0.397
HadCRUT4	0.022	0.015	0.030	-0.005	0.012	0.026	0.000	0.022
ADS	0.005	-0.027*	0.028*	0.002	-0.002	-0.004	-0.007	-0.007
USEPU	0.017	-0.004	-0.003	0.026	-0.001	0.004	0.029	0.030
VIX	-0.024	-0.013	0.002	0.022	0.030	0.038	-0.029	0.033
OILV	-0.389	-0.204	-0.174	-0.613**	-0.490	-0.722	-0.551	-0.049
S&P 500(-1)	0.177	0.141	0.032	-0.088	-0.108	-0.350*	0.363	0.164
USD(-1)	-0.174	-0.649**	-0.517*	-0.097	-0.371	-0.405	0.026	-0.471
ADS(-1)	-0.011	-0.003	-0.035**	-0.017	-0.027	-0.011	-0.016	0.024
HadCRUT4(-1)	-0.010	-0.020	-0.015	-0.027	-0.030	0.001	-0.007	-0.041
USEPU(-1)	0.009	0.015	0.002	-0.013	-0.006	-0.040	-0.024	-0.010
VIX(-1)	0.048	0.007	0.020	0.015	0.017	-0.020	0.072*	0.044
OILV(-1)	-0.093	-0.053	-0.321	-0.029	-0.190	-0.046	0.141	-0.397
Crisis dummy	0.000	-0.004	0.002	0.002	-0.002	0.003	0.002	-0.001
Constant	0.005	0.006	0.005	0.006	0.009	0.009	0.002	0.003

Notes. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

one subsequent month as their one-period lagged coefficients are statistically significant. This result may indicate that investors in agricultural commodity futures possibly underreact to changes in the USD, resulting in delayed incorporation of information. This finding is also consistent with the gradual information diffusion hypothesis proposed in Hong and Stein (1999).

All remaining one-period lagged variables' effect are statistically insignificant, except for the negative impact of ADS on rapeseed and borderline significant impact of VIX on cotton futures returns. The crisis dummy variable is insignificant in all cases, indicating that the food-price crisis of 2006 did not significantly impact the agricultural commodity futures market.

5.2. Quantile regression (QR) results

This subsection presents the QR results involving agricultural commodity futures returns, temperature anomaly, and macroeconomic fundamentals. Table 5 demonstrates results related to soybean futures. Regarding the S&P500, the coefficients are statistically insignificant across the quantiles. However, the one-period lagged S&P500 coefficient is borderline significant in the extreme low ($\tau = 0.05$) quantile. These results indicate that, overall, stock market returns have a weak or no influence on soybean futures returns. Although the notion of

financialization states that increased investments in agricultural commodities increase spillovers from equities to commodities, our result does not support this rationale. The statistically insignificant relationship between the S&P500 and soybean futures returns also contradicts the findings reported in Silvennoinen and Thorp (2013). Overall, our results indicate that soybean futures may be considered an effective hedging tool in a portfolio based on equities.

A significant negative relationship is found between changes in the relative strength of the USD and soybean future returns during moderately bad, normal, and moderately good market conditions ($\tau = 0.25$ to $\tau = 0.75$). This finding implies that the increasing strength of the USD may induce a decrease in the cost of harvesting, increasing the supply and further driving soybean prices downwards. However, this result is instantaneous as the lagged effect of the USD is statistically insignificant irrespective of the quantiles. In general, our findings first indicate that soybean may be considered as an appropriate tool for hedging a USD-weighted portfolio. Secondly, it supports the results reported in Nazliglu and Soytaş (2012).

Temperature anomaly exhibits a weak positive impact on soybean futures returns during exceptionally good market conditions ($\tau = 0.95$), implying that an increase in temperature anomaly contributes to an increase in soybean futures returns when the market is bullish. This result may be linked to the basic demand-supply price dynamics. An

Table 5

Quantile regression results: soybean.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	-0.019	-0.007	-0.023	-0.003	-0.072	-0.045	-0.026
S&P 500	0.427	0.410	0.338	0.166	0.318	0.057	0.286
USD	0.447	-0.339	-0.852*	-0.856**	-1.064**	-0.987	-0.805
HadCRUT4	-0.021	0.002	0.013	0.018	0.019	0.064	0.078*
ADS	-0.001	0.014	0.005	0.009	-0.012	-0.023	0.005
USEPU	0.050	0.050	0.018	0.040	0.055	-0.029	0.018
VIX	-0.008	-0.013	-0.055	-0.030	-0.011	-0.063	-0.006
OILV	0.040	-0.032	-0.188	-0.463	-0.324	-0.784	-0.469
S&P 500(-1)	0.665*	0.297	0.316	0.212	0.074	0.053	-0.009
USD(-1)	-0.865	-0.304	-0.022	-0.189	-0.595	-0.162	-0.226
ADS(-1)	0.008	-0.008	-0.023	-0.021	-0.011	0.039	0.050
HadCRUT4(-1)	-0.127**	-0.089*	-0.029	-0.007	0.022	0.070	0.076
USEPU(-1)	-0.031	-0.023	-0.044	0.033	0.053	0.027	0.087
VIX(-1)	0.246***	0.117**	0.081*	0.032	-0.004	0.059	0.011
OILV(-1)	0.591	0.478	0.111	-0.004	-0.356	-0.365	-0.156
Crisis dummy	-0.005	0.003	0.001	-0.001	-0.007	-0.008	-0.012
Constant	-0.111***	-0.088***	-0.042***	0.005	0.050***	0.110***	0.128***
Pseudo-R ²	0.178	0.157	0.092	0.089	0.100	0.122	0.190

Notes. Quantile regression results of soybean futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

increase in the temperature anomaly decreases soybean supply without a corresponding decrease in demand, which consequently drives the price upwards.

We further observe that lagged temperature anomaly has a negative impact on soybean futures returns in low quantiles ($\tau = 0.05$ and $\tau = 0.10$). This result suggests that during bad market conditions, an increase in temperature anomaly may be taken as an indication of increased uncertainty, bringing a negative impact on the one-period-ahead soybean futures returns. These findings further imply that soybean futures returns show asymmetric response to the change in temperature anomaly depending on the state of the market.

Table 6 presents results exploring sunflower futures returns. The S&P500 return is found to have a significant positive impact on sunflower futures returns at low ($\tau = 0.25$) and high quantiles ($\tau = 0.90$) but not in the middle quantile. This result suggests that during moderately bad and good conditions in the sunflower futures market, increasing stock market returns generates larger sunflower futures returns. We further find that the lagged stock market returns have a marginally significant positive impact on sunflower futures returns during good ($\tau = 0.75$ and $\tau = 0.95$) market conditions for sunflower futures. These results, firstly, may be attributed to the increased financialization of the commodities market, as highlighted in Creti et al. (2013) and, secondly, imply that sunflower futures may provide diversification benefits for stock market investors during normal conditions in the sunflower futures market. Nonetheless, this hedging benefit may disappear during moderately bad and good market conditions. The lagged effect further indicates a gradual incorporation of equity market information on the agricultural commodity markets.

In accordance with the results found for soybean futures, sunflower futures returns respond negatively to the changes in the relative strength of the USD during moderately bad ($\tau = 0.25$) to exceptionally good ($\tau = 0.95$) sunflower market conditions. The lagged effect of the USD is also negative and statistically significant in the extreme low quantile ($\tau = 0.05$). These results suggest that the USD provides diversification benefits for commodity (in this particular case, sunflower) investors and vice versa.

Furthermore, investors holding an USD-weighted portfolio should be aware of price fluctuations in the sunflower futures from fluctuations in the USD. Our results, in general, support the findings presented in Nazlioglu and Soytaş (2012), showing a strong relationship between exchange rates and agricultural commodity prices. In addition, temperature anomaly has a statistically insignificant impact on the contemporaneous and one-period-ahead sunflower returns.

Table 6
Quantile regression results: sunflower.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	-0.153	-0.114	-0.044	0.088	0.105	-0.007	-0.259
S&P 500	-0.008	0.375	0.412**	0.272	0.301	0.625**	0.442
USD	-0.742	-0.299	-0.835*	-1.020***	-1.413***	-1.597***	-2.268***
HadCRUT4	0.017	-0.043	-0.003	0.010	-0.017	-0.009	0.085
ADS	-0.025	-0.035*	-0.055***	-0.018	-0.018	-0.019	-0.041
USEPU	0.033	-0.013	-0.032	-0.005	-0.031	0.038	0.036
VIX	-0.056	0.004	0.045	0.006	0.007	0.022	-0.036
OILV	0.259	-0.460	-0.104	-0.253	0.359	-0.272	-0.678
S&P 500(-1)	-0.262	-0.041	0.024	0.141	0.357*	0.226	0.697*
USD(-1)	-1.223*	-0.338	-0.497	-0.660	-0.586	-0.428	-0.499
ADS(-1)	0.023	0.031	0.031	-0.012	-0.031	-0.042	-0.050
HadCRUT4(-1)	-0.044	-0.038	-0.018	-0.020	-0.003	-0.010	0.037
USEPU(-1)	-0.046	0.027	0.027	0.003	0.018	0.068	0.007
VIX(-1)	-0.122	-0.042	-0.031	0.008	0.056	0.006	0.069
OILV(-1)	-0.330	-0.107	-0.244	0.039	-0.163	0.370	-0.213
Crisis dummy	0.043**	0.016	0.016	0.005	-0.024	-0.035	-0.034
Constant	-0.131***	-0.082***	-0.051***	-0.003	0.056***	0.098***	0.137***
Pseudo-R ²	0.228	0.163	0.125	0.098	0.150	0.204	0.234

Notes. Quantile regression results of sunflower futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

The results related to rapeseed futures are reported in Table 7. We observe that the aggregate stock market interacts with the rapeseed futures market during extremely bad ($\tau = 0.05$) and bad ($\tau = 0.25$) conditions on the rapeseed futures market. We find a statistically significant coefficient of the S&P500 return for such quantiles. This relationship, however, is concurrent as the coefficient of the lagged stock market returns is statistically insignificant. These results imply that rapeseed futures provide diversification benefits for stock market investors during normal and good conditions in the rapeseed market. However, when the rapeseed futures market becomes bearish, it then becomes positively related to the stock market and, consequently, diversification benefit declines or even disappears.

The rapeseed futures returns exhibit a negative response to the relative strength of the USD across the quantiles, except for the cases of $\tau = 0.75$ and $\tau = 0.95$. However, the impact of the USD on the rapeseed futures returns weakens when we consider lagged USD - the coefficient is significant only at $\tau = 0.75$. Firstly, this result is in accordance with Nazlioglu and Soytaş (2012) and, secondly, it implies possibilities to hedge a USD-weighted portfolio by including agricultural commodity futures. Moreover, no temperature anomaly effect on the rapeseed futures returns may be observed irrespective of market conditions, suggesting that rapeseed futures returns are determined by macroeconomic fundamentals rather than temperature anomaly.

Table 8 reports results involving corn futures. Consistent with our previous results, the S&P500 return positively influences corn futures returns during bearish market conditions ($\tau = 0.05$ to $\tau = 0.25$). As previously reasoned, this result may arise since when the S&P500 return changes, then there is a spillover effect and the corn futures returns show a movement in a similar direction. This result is only observed during a bearish condition in the corn futures market. Once again, this result manifests the presence of hedging possibilities between the aggregate stock market and corn futures while the corn market is in bullish condition. However, the hedging possibilities dissipate when the corn market enters in a bearish condition. The impact of the S&P500 return on corn futures is only contemporaneous as the coefficient of the lagged S&P500 return is statistically insignificant irrespective of the quantiles.

The temperature anomaly effect on the corn futures returns is observed during exceptionally bad ($\tau = 0.05$), bad ($\tau = 0.10$), and exceptionally good ($\tau = 0.95$) market conditions. Interestingly, the coefficients are negative in low quantiles, while the coefficient is positive in the extreme high quantile. These results represent that when the returns on corn futures are at low levels, an increase in the temperature anomaly decreases corn futures returns. However, when the corn futures

Table 7
Quantile regression results: rapeseed.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	0.037	-0.050	0.048	0.044	0.030	0.098	0.145
S&P 500	0.583*	0.670**	0.265*	0.026	0.020	-0.063	-0.252
USD	-1.171**	-0.891**	-0.897**	-0.820**	-0.534	-0.863**	-0.684
HadCRUT4	0.005	0.031	0.038	0.005	0.027	0.029	0.027
ADS	-0.002	0.018	0.017	0.016	0.040**	0.049*	0.073**
USEPU	0.008	-0.046	-0.030	-0.012	-0.006	0.012	0.008
VIX	0.122**	0.058	0.043	-0.011	-0.025	-0.067	-0.111**
OILV	0.051	-0.179	-0.538	-0.400	-0.207	0.552	0.466
S&P 500(-1)	0.368	0.300	-0.073	-0.010	0.032	-0.069	-0.032
USD(-1)	0.246	0.261	-0.368	-0.491	-0.918***	-0.487	-0.153
ADS(-1)	-0.005	-0.032	-0.024	-0.039**	-0.038*	-0.033	-0.061*
HadCRUT4(-1)	-0.047	-0.022	0.002	-0.019	-0.032	-0.033	-0.040
USEPU(-1)	0.009	-0.042	-0.013	-0.009	-0.018	0.013	0.027
VIX(-1)	0.066	0.083*	0.035	0.014	0.031	-0.007	-0.065
OILV(-1)	0.002	-0.297	-0.548	-0.320	-0.234	-0.414	-0.473
Crisis dummy	0.004	-0.001	0.000	-0.001	0.000	0.010	0.016
Constant	-0.089***	-0.061***	-0.018	0.013	0.038***	0.055***	0.068***
Pseudo-R ²	0.230	0.180	0.109	0.129	0.180	0.251	0.306

Notes. Quantile regression results of rapeseed futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

Table 8
Quantile regression results: corn.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	-0.057	0.003	0.005	0.024	0.114	0.026	0.297*
S&P 500	1.065***	1.038***	0.559**	0.333*	0.092	0.298	-0.183
USD	-0.416	-0.144	-0.743	-0.559	-0.100	-0.582	-0.253
HadCRUT4	-0.096*	-0.120***	-0.042	-0.012	0.004	0.054	0.085*
ADS	0.014	0.009	0.015	-0.011	-0.031	-0.054	-0.093
USEPU	0.022	0.002	0.006	0.049	0.051	-0.031	-0.110
VIX	0.125*	0.099	0.051	-0.001	-0.041	0.014	0.050
OILV	0.081	0.193	-0.631	-0.692	-0.073	-0.728	-0.343
S&P 500(-1)	0.123	-0.009	-0.064	-0.011	0.001	-0.205	-0.388
USD(-1)	0.953	0.360	0.235	0.033	-0.972**	-0.293	-0.202
ADS(-1)	-0.008	-0.038	-0.029	-0.016	0.015	0.063	0.033
HadCRUT4(-1)	-0.041	-0.064	-0.015	-0.038	-0.014	-0.024	0.029
USEPU(-1)	-0.079*	-0.069*	-0.028	0.020	-0.002	0.035	0.055
VIX(-1)	0.013	-0.003	0.007	0.042	-0.008	-0.042	0.102
OILV(-1)	0.311	0.258	-0.129	0.182	-0.226	1.337	1.464
Crisis dummy	-0.028	-0.002	-0.004	0.003	0.000	0.018	0.061**
Constant	-0.094***	-0.092***	-0.034***	0.002	0.045***	0.068***	0.093***
Pseudo-R ²	0.240	0.187	0.118	0.070	0.061	0.117	0.184

Notes. Quantile regression results of corn futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

returns are at extreme high levels, an increase in the temperature anomaly increases corn futures returns. As indicated previously, this result implies that the direction of temperature effect on the agricultural commodity futures returns is conditional on the commodity futures market condition.

Although we previously found some statistically significant impact of the relative strength in the USD on agricultural commodity futures returns, we find that this variable does not have any explanatory power over fluctuations of corn futures returns. This result holds for the contemporaneous and lagged effect except for the statistically significant coefficient of lagged USD in moderately high quantile ($\tau = 0.75$). This result implies that corn futures may provide important diversification benefits for investors holding a USD-weighted portfolio.

Table 9 demonstrates the QR results involving wheat futures return as the dependent variable. In accordance with the relationship between other agricultural commodity futures and the S&P500 returns, we find that the latter has a statistically significant positive impact on wheat futures returns during exceptionally bad ($\tau = 0.05$) to normal ($\tau = 0.50$) conditions in the wheat market. This finding suggests a dependence between both asset classes, especially when the agricultural commodity

market condition is in bad to normal conditions. However, this dependence dissipates when the wheat market is bullish. Furthermore, as previously highlighted, this result implies that wheat futures provide limited hedging opportunities for equity market investors since they increasingly become interconnected when the wheat market is bearish to normal. However, wheat futures and equity may be included in the same portfolio for diversification purposes when the wheat market is bullish. Nonetheless, the positive effect of the aggregate stock returns on the wheat futures returns is contemporaneous as the coefficient of the lagged independent variable is statistically insignificant across the quantiles.

In contrast to the results reported for most of the other agricultural commodity futures, the relative strength of the USD does not appear to have any explanatory power for wheat futures returns. This result holds for both contemporaneous and lagged effect. This result implies that wheat futures have a certain characteristic, distinct from other agricultural commodity futures, insensitivity to USD fluctuations. The temperature anomaly, in general, does not exhibit any power to explain wheat futures returns. Nevertheless, the coefficient of temperature anomaly is marginally significant in the high ($\tau = 0.90$) quantile,

Table 9
Quantile regression results: wheat.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	-0.282**	-0.227**	-0.149	0.020	0.039	0.287*	0.508**
S&P 500	0.523*	0.814***	0.516**	0.361*	0.051	0.327	0.063
USD	1.084	-0.280	-0.261	-0.245	-0.069	0.120	0.894
HadCRUT4	0.008	-0.016	0.010	0.018	0.033	0.075*	0.072
ADS	-0.020	-0.018	-0.003	0.002	0.022	-0.045	0.035
USEPU	-0.002	-0.015	-0.020	-0.006	-0.006	0.019	0.031
VIX	0.042	0.094	0.070	0.047	-0.028	-0.043	-0.075
OILV	-0.604	-1.272*	-0.457	-0.515	-0.185	-0.536	-0.641
S&P 500(-1)	-0.195	-0.492	-0.111	-0.052	-0.015	-0.279	-0.391
USD(-1)	-0.183	0.424	0.433	-0.413	-0.274	-0.810	-1.069
ADS(-1)	-0.006	-0.021	-0.011	-0.027	-0.021	0.003	0.020
HadCRUT4(-1)	0.013	0.013	-0.009	-0.034	-0.017	-0.023	-0.068
USEPU(-1)	-0.118*	-0.059	-0.020	0.014	0.003	0.120	0.099
VIX(-1)	-0.003	-0.041	-0.010	0.027	0.043	0.033	0.010
OILV(-1)	-0.506	-0.717	0.029	0.069	-0.132	0.651	0.553
Crisis dummy	-0.052**	-0.041**	-0.021	-0.003	0.027*	0.020	0.031
Constant	-0.074***	-0.044***	-0.028**	0.006	0.034**	0.085***	0.107***
Pseudo-R ²	0.276	0.206	0.075	0.058	0.063	0.101	0.171

Notes. Quantile regression results of wheat futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

suggesting that an increase in the temperature anomaly is positively related to the wheat futures returns during good market conditions.

Table 10 reports results regarding cocoa futures. Cocoa futures returns appear to be invariant to the changes in the S&P500 stock market prices and the relative strength of the USD. These results are at odds with the relationship between agricultural commodity futures and the S&P500 returns as well as the USD previously reported in this subsection. This finding, however, is consistent with Nazlioglu and Soytaş (2012). Nevertheless, we find a weak positive relationship between the S&P500 and cocoa futures returns during bearish condition in the cocoa market ($\tau = 0.10$). This result implies that cocoa futures provide excellent hedging opportunities for equity investors while the cocoa futures market is normal to bullish. Hedging opportunity, however, is slightly limited when the cocoa market is in a down condition. We further find a weak negative impact of the S&P500 returns on the one-period-ahead of cocoa futures returns when the cocoa market is moderately bearish. This result indicates that when stock return is upward, investors shift their investment from cocoa futures to the stock market, leading to a decline in cocoa futures returns in the subsequent period. This result, however, is marginally significant.

Table 10
Quantile regression results: cocoa.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	-0.666***	-0.519***	-0.341***	-0.150	-0.130	-0.230**	-0.302**
S&P 500	0.176	0.480*	0.477	0.214	-0.145	-0.458	0.168
USD	0.121	-0.777	-0.316	-0.123	-0.665	-0.460	-0.837
HadCRUT4	-0.002	0.047	-0.010	-0.029	0.024	0.043	0.092*
ADS	0.028	0.016	-0.012	0.001	0.015	-0.009	0.003
USEPU	0.097	0.072	0.003	-0.042	0.012	0.057	-0.025
VIX	0.005	0.067	0.089	0.022	-0.018	-0.063	0.028
OILV	-1.031	-0.607	-0.912	-0.975	-0.522	-0.077	-0.613
S&P 500(-1)	-0.107	-0.207	-0.483*	-0.477	-0.425	-0.339	-0.126
USD(-1)	-0.294	-0.411	-0.265	-0.380	-0.529	-1.464**	-0.507
ADS(-1)	-0.057	-0.021	0.010	0.011	-0.018	-0.027	-0.035
HadCRUT4(-1)	-0.011	-0.021	-0.013	-0.038	-0.038	0.029	0.054
USEPU(-1)	-0.063	-0.005	-0.033	-0.058	-0.068	-0.026	-0.014
VIX(-1)	0.081	0.025	-0.023	-0.063	-0.037	0.014	0.045
OILV(-1)	-0.298	-0.194	-0.605	-0.177	-0.190	-0.763	0.330
Crisis dummy	0.020	0.018	0.017	0.022	0.005	0.002	-0.039
Constant	-0.129***	-0.094***	-0.047***	-0.002	0.056***	0.109***	0.156***
Pseudo-R ²	0.278	0.204	0.142	0.086	0.095	0.126	0.169

Notes. Quantile regression results of cocoa futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

The significant negative impact of the lagged USD on cocoa futures returns further indicates that commodity investors should be aware of the sensitivity of commodity prices to changes in the USD exchange rates. We also find that an increase in the temperature anomaly has a weak positive effect on the cocoa futures returns during exceptionally good market conditions ($\tau = 0.95$). However, cocoa futures returns are insensitive to the temperature anomaly when they are in normal to bearish states. This result is mostly in accordance with the results presented for other agricultural commodity futures.

Table 11 reports results involving cotton futures. Consistent with the relationship between the S&P500 returns and agricultural commodity futures returns identified for most of the agricultural commodity futures, we find that stock market returns have a significant positive impact on both the current and one-period-ahead cotton futures returns when the cotton market is extremely bearish ($\tau = 0.05$ to $\tau = 0.10$). However, this relationship is not evident when the cotton market is in a normal to bullish condition. As previously discussed, this result, firstly, indicates spillover from equity to the agricultural commodity market and, secondly, implies limited hedging opportunities particularly when the cotton market is bearish. However, statistically insignificant coefficients

Table 11
Quantile regression results: cotton.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	-0.291*	-0.173	-0.061	-0.187*	-0.130	-0.197	-0.159
S&P 500	0.458	0.737**	0.478	0.242	0.057	0.112	0.350
USD	-1.675**	-1.339**	-1.017*	-0.706*	-1.028**	-1.342**	-1.676**
HadCRUT4	-0.122**	-0.091**	-0.030	-0.006	0.014	0.026	0.027
ADS	-0.046	0.003	-0.024	-0.008	-0.002	-0.036	-0.047
USEPU	0.035	0.088	0.066*	0.021	0.000	-0.030	-0.070
VIX	-0.072	-0.041	-0.014	-0.047	-0.047	-0.082	-0.010
OILV	-0.501	-0.368	-0.702	-0.164	0.126	-0.369	-0.464
S&P 500(-1)	0.853**	1.015***	0.449	0.269	0.088	0.321	0.319
USD(-1)	1.241	1.130	0.354	-0.252	-0.186	-0.476	-0.398
ADS(-1)	-0.058	-0.048	-0.019	-0.022	-0.020	0.017	0.007
HadCRUT4(-1)	-0.124**	-0.094**	-0.027	0.015	-0.005	0.069*	0.072
USEPU(-1)	0.003	0.043	-0.005	0.021	-0.022	-0.014	-0.040
VIX(-1)	0.180**	0.187***	0.091*	0.042	0.030	-0.043	-0.013
OILV(-1)	0.167	-0.167	-0.119	0.130	-0.030	-0.055	-0.103
Crisis dummy	-0.020	0.005	0.022	0.015	0.006	-0.023	-0.017
Constant	-0.106***	-0.096***	-0.054***	-0.008	0.037**	0.116***	0.124***
Pseudo-R ²	0.280	0.221	0.124	0.084	0.085	0.144	0.205

Notes. Quantile regression results of cotton futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

for normal to bullish conditions indicate that the cotton market may still provide diversification benefits for equity investors during such conditions.

It is found that the relative strength of the USD has a significant negative impact on cotton futures returns. This result holds for all quantiles. This finding supports the notion of financialization that interconnects the commodity futures and currency markets. This result implies that by adding cotton futures in a USD-weighted portfolio, an investor may safeguard against major fluctuations in the relative strength of the USD. This benefit may be achieved regardless of the condition of the cotton futures market. However, the negative impact of the relative strength of the USD on cotton futures returns is a short-run phenomenon as the coefficient of the lagged variable is statistically insignificant across the quantiles.

Both the lagged and current temperature anomaly are found to have a significant negative impact on current cotton futures returns when the cotton futures market is extremely bearish ($\tau = 0.05$ to $\tau = 0.10$). However, this result is not found when the cotton market is under normal condition. Although the temperature anomaly does not have a contemporaneous effect on cotton futures returns in bullish market

condition, the temperature anomaly has a weak positive impact on the one-period-ahead cotton futures returns.

Overall, our result implies that an increase in the temperature anomaly manifests increased uncertainty in the agricultural commodity market, ultimately leading to a contemporaneous decline in the returns. This effect is observed when the commodity market is already in a down condition. Our finding may further indicate that commodity investors tend to ignore the impact of temperature anomaly during normal to bullish market conditions while they become more risk-averse and respond negatively during a bearish market condition.

Table 12 presents results related to coffee futures returns. Consistent with our previous results for most agricultural commodity futures, the S&P500 returns seem to have a significant positive impact on coffee futures returns when the coffee market is extremely bearish. Nonetheless, this positive impact is short-term as the S&P500 returns may only explain the contemporaneous but not the one-month-ahead coffee futures returns. These results are in accordance with Tang and Xiong (2012), who show a positive return correlation between non-energy commodities and the S&P500 stock returns, particularly after 2008. Further, our results support Lombardi and Ravazzolo (2016), who find

Table 12
Quantile regression results: coffee.

	$\tau = 0.05$	$\tau = 0.10$	$\tau = 0.25$	$\tau = 0.50$	$\tau = 0.75$	$\tau = 0.90$	$\tau = 0.95$
Lagged dependent variable	-0.429***	-0.363***	-0.291***	-0.315***	-0.157	-0.098	0.104
S&P 500	0.774***	0.619**	0.325	0.276	0.153	-0.551	-0.590
USD	-1.461**	-0.521	-0.265	-0.391	-0.831	-0.924	-0.832
HadCRUT4	-0.013	-0.009	-0.001	0.046	0.026	0.044	0.024
ADS	-0.058**	-0.013	-0.038	-0.016	0.000	-0.013	0.029
USEPU	0.037	0.082*	0.004	0.028	0.026	-0.027	0.036
VIX	0.090	0.041	0.028	0.020	-0.036	-0.012	-0.062
OILV	0.104	-0.203	-0.014	-0.101	-0.159	0.533	0.055
S&P 500(-1)	0.358	0.318	0.122	0.339	0.055	0.001	-0.371
USD(-1)	0.102	0.032	-0.103	-0.463	-1.006	-1.333	-2.237**
ADS(-1)	0.012	0.002	0.010	0.019	0.016	0.024	-0.018
HadCRUT4(-1)	-0.030	-0.057	-0.007	-0.004	-0.051	-0.111*	-0.185***
USEPU(-1)	0.052	0.047	-0.012	-0.020	-0.006	0.021	-0.009
VIX(-1)	0.113*	0.095	0.013	0.025	-0.001	0.076	0.097
OILV(-1)	0.150	-0.163	-0.291	-0.349	-0.271	-1.090	-1.595**
Crisis dummy	0.005	0.001	0.008	0.019	-0.011	-0.009	-0.039
Constant	-0.128***	-0.094***	-0.058***	-0.025	0.062***	0.131***	0.201***
Pseudo-R ²	0.309	0.218	0.142	0.085	0.083	0.139	0.214

Notes. Quantile regression results of coffee futures returns regressed on the temperature anomaly and macroeconomic fundamentals. Standard errors are estimated using the design matrix bootstrapping procedure. See notes to Table 2 for the definitions of the variables. ***, **, * respectively indicate statistical significance at the 1%, 5%, and 10% level.

an increasing correlation between the global stock market and commodities after the GFC of 2007–2008. Moreover, the positive relationship implies limited hedging opportunities involving agricultural commodity futures for equity portfolios as they exhibit a co-movement, particularly during bearish condition in the commodity market.

Consistent with the results presented for cotton, rapeseed, soybean, and sunflower futures, the relative strength of the USD exerts a significant negative impact on coffee futures returns when the coffee futures return is in the extreme low quantile ($\tau = 0.05$). Nevertheless, this result is not observed during normal to a bullish condition in the coffee market. Our finding is consistent with [Nazlioglu and Soytaş \(2012\)](#), who report a negative relationship between the USD and agricultural commodity prices.

We also find evidence of a negative relationship between current coffee futures returns and one-period-lagged USD when the coffee futures returns are in the extreme high quantile ($\tau = 0.95$). This result, firstly, suggests a negative dynamic relationship between the USD and coffee futures and, secondly, indicates hedging opportunities between currency and commodity investments. However, the relationship between the USD and coffee futures returns is not as strong as the relationship between soybean, sunflower, corn, and cotton futures with the USD.

Although the temperature anomaly does not significantly impact the contemporaneous coffee futures returns, we observe that the temperature anomaly exhibits a significant negative impact on the one-period-ahead coffee futures returns when the coffee market is extremely bullish ($\tau = 0.90$ to $\tau = 0.95$). As previously discussed, this result may indicate that an increase in temperature anomaly increases uncertainty in the commodity market, which ultimately reduces coffee futures returns. The one-period lagged effect may be attributable to the fact that commodity investors linger assessing the impact of temperature anomaly on coffee futures, which leads to a gradual spread of information.

Regarding the effect of lagged dependent variable, we find that in the cases of soybean, sunflower, rapeseed, and corn futures, there is no dynamic relationship between past and current returns as the coefficient of the lagged dependent variables is statistically insignificant across the quantiles. This result indicates that past returns are unlikely to improve future returns forecasting for these agricultural commodity futures. We obtain different results for wheat, cocoa, cotton, and coffee futures. For instance, it is found that the lagged dependent variable has a statistically significant negative impact on current returns, when the wheat ([Table 9](#)) and cocoa ([Table 10](#)) futures markets are in $\tau = 0.05$ to $\tau = 0.10$ and $\tau = 0.05$ to $\tau = 0.25$ quantiles, respectively. This result implies that in such instances, although the current returns cannot predict the one-period-ahead returns in the normal market condition, during a bearish condition, the past returns negatively predict future returns. This result further indicates that these futures returns exhibit a mean reversion property in a down market condition. As such, a negative return predicts an one-period-ahead positive return. The negative impact of the lagged dependent variable for cotton ([Table 11](#)) and coffee ([Table 12](#)) is found for the $\tau = 0.05$ to $\tau = 0.50$ quantiles, indicating that futures returns become highly correlated with their past returns during bearish to normal conditions. However, during a bullish market condition, information embedded in past returns is immediately reflected. Therefore, past returns do not carry information to future returns. Lastly, in high quantiles ($\tau = 0.90$ to $\tau = 0.95$), for wheat and cocoa futures, the lagged dependent variable is found to have a statistically significant impact on current returns. This result manifests a dynamic relationship when the wheat and cocoa futures markets are in exceptionally good condition.

We do not find any contemporaneous or lagged effect of ADS on soybean, corn, wheat, cocoa, and cotton futures returns, implying that these futures returns are invariant to the variations in economic activities. However, sunflower ([Table 6](#)) and coffee ([Table 12](#)) futures returns negatively respond to ADS during bad market conditions ($\tau = 0.10$ to $\tau = 0.25$). A possible reason for this negative relationship may be that when economic condition strengthens but the futures markets are still

bearish, investors may shift their investment to more cyclical assets (e.g., equity), causing a downward trend in the futures markets. This result, however, is contemporaneous, as the lagged effect of ADS on sunflower futures returns is found to be statistically insignificant across the quantiles. Interestingly, in the case of rapeseed futures returns, ADS has a positive impact during good market conditions ($\tau = 0.75$ to $\tau = 0.95$), as reported in [Table 7](#). This result implies that the direction of the relationship of ADS with agricultural commodity futures may be different based on a particular agricultural commodity market condition. Nonetheless, the lagged effect of ADS on the rapeseed futures market is negative and statistically significant in the middle to high quantiles ($\tau = 0.50$, $\tau = 0.75$, and $\tau = 0.95$). These results may arise when the economic condition strengthens, and the rapeseed market tends to be strong, investors' positive expectation drives the return further. On the other hand, investors' profit-taking behavior may lead to a decline in rapeseed futures returns when both ADS and the rapeseed market are strong.

In the cases of soybean, sunflower, rapeseed, corn, cocoa, and cotton futures, we do not find any contemporaneous or lagged effect of EPU and OILV, implying that these futures returns are invariant to uncertainties related to economic policy and the crude oil market. Overall, our results indicate that these futures markets are mostly segmented from other markets and do not respond to changes in macroeconomic fundamentals. Our result is in accordance with [Caporale et al. \(2017\)](#), that find statistically insignificant impact of macroeconomic news on soybean futures returns. However, our finding contradicts [Natanelov et al. \(2011\)](#), who report a co-movement between crude oil and soybean. The discrepancy may arise as we use crude oil price volatility instead of returns.

Among the uncertainty variables, the VIX appears to have some impact on agricultural commodity futures returns. For instance, we find that the lagged VIX has a significant positive impact in low quantiles ($\tau = 0.05$ to $\tau = 0.10$) of soybean, rapeseed, corn, cotton, and coffee futures returns. This result indicates that an increase in stock market volatility increases one-period-ahead futures returns when these futures markets are bearish. This finding may be linked with the idea that when investors anticipate higher volatility in the equity market, due to their flight-to-safety preference, investors shift their investment from equity to commodity markets, which increases commodity futures returns in the subsequent period. This finding further indicates important hedging possibilities for equity investors arising from agricultural commodity futures. We also find that the VIX has a significant positive (negative) impact on rapeseed futures return in the extreme low (high) quantile, indicating equity market uncertainty's asymmetric impact across quantiles ([Table 7](#)). This result is in accordance with the effect of the VIX on rapeseed futures returns and the results reported in [Creti et al. \(2013\)](#).

The EPU has a marginally significant positive impact on coffee futures in low quantiles ($\tau = 0.05$ to $\tau = 0.10$), as reported in [Table 12](#). This result may indicate that an increase in economic policy uncertainty may lead investors to shift their funds from more cyclical assets to agricultural commodities (e.g., coffee), causing an increase in coffee futures returns. Since an increase in the EPU may lead to an increase in coffee futures returns, hedging possibilities may occur in terms of coffee futures used as a protection against increases in macroeconomic uncertainty. This finding is in congruence with [Joëts et al. \(2017\)](#) and consistent with the results related to corn futures.

In the case of wheat futures, OILV has a marginally significant negative impact on the low ($\tau = 0.10$) quantile, indicating that an increase in OILV would decrease wheat futures returns ([Table 9](#)). Moreover, lagged OILV presents a statistically significant negative impact on coffee futures returns when the coffee market is extremely bullish ($\tau = 0.95$) ([Table 12](#)). These results indicate that an increase in OILV is perceived negatively by agricultural commodity investors, leading to a decline in commodity returns. The lagged effect is consistent with the gradual information diffusion hypothesis.

Consistent with our OLS results, the crisis dummy is statistically insignificant across the quantiles for soybean, rapeseed, coffee, cocoa, and cotton futures returns, indicating that these futures returns remain unaffected by the food-price crisis of 2006. However, in the cases of sunflower (Table 6) and corn (Table 8) futures returns, the crisis dummy is positive and statistically significant in the extreme low ($\tau = 0.05$) and extreme high ($\tau = 0.95$) quantiles, respectively. This result indicates that a crisis, in general, further increases returns when these futures markets are in a particular quantile. This finding may be attributed to the notion that investors tend to shift their investment to these futures markets during a crisis period due to their hedging characteristic that further drives up futures prices.

6. Conclusion

The present study examines if temperature anomaly and macroeconomic fundamentals explain conditional quantiles of agricultural commodity futures returns. This analysis is worthwhile in the context of frequent extreme weather events and their potential link with higher volatility in the agricultural commodity markets, and lack of empirical research on the impact of temperature anomaly and macroeconomic fundamentals on agricultural commodity futures returns under different market conditions, such as bearish, normal, and bullish.

We consider returns from two asset classes and changes in economic activity, temperature anomaly, and different uncertainty measures to explain eight agricultural commodity futures returns. A sample consisting of monthly data frequency from January 1999 to December 2018 is considered. We use the QR modeling approach that, firstly, enables us to examine the co-movement and causal relationship and, secondly, allows us to investigate the relationship under distinct market conditions. Moreover, we examine both the contemporaneous and dynamic relationship between agricultural commodity futures prices and the explanatory variables.

We find that temperature anomaly has a significant negative (positive) impact on returns of soybean, corn, cotton, and coffee (soybean, corn, and cocoa) futures in extreme bearish ($\tau = 0.05$) and bullish ($\tau = 0.95$) market conditions. This indicates that investors value temperature anomaly information, particularly under extreme market conditions. We further show a significant positive relationship between agricultural commodity futures returns - in the cases of corn, rapeseed, wheat, cotton, and coffee - and the S&P500 return in low quantiles, typically between $\tau = 0.05$ and $\tau = 0.10$. Although agricultural commodity futures returns are found to respond negatively to exchange rate changes, this result varies across commodities and quantiles. Moreover, changes in economic activity, uncertainties associated with economic policy, equity market, and oil prices typically do not have a strong impact on agricultural commodity futures returns.

Appendix

Our results have several implications regarding portfolio construction, investment diversification, and hedging strategies. A positive relationship between the equity and agricultural commodity futures market in low quantiles implies that they may be used for hedging or diversification purposes under normal conditions. However, the diversification benefits may decline or even disappear under a bearish market condition. Agricultural commodity futures may also be used to hedge against USD exchange rate fluctuations. Therefore, while assessing and investing in agricultural commodity futures, investors should incorporate temperature anomaly information in their decision-making process, particularly under extreme market conditions. Due to the prevalence of larger levels of uncertainty due to a greater frequency of extreme weather events, our study suggests that an adjustment in terms of risk evaluation of weather shocks (or temperature in general) would benefit investment portfolio management. For instance, Engle et al., 2020 climate-risk-hedged portfolio, although underperform in the short-run compared to Markowitz mean-variance efficient portfolios, exhibits superior performance in the long-run through compensating the losses arising from climate events.

A limitation of the present study is that the results derived from the adopted approach (i.e., QR) are not compared with that of the competing frameworks. We therefore cannot empirically confirm the superiority of the chosen modeling approach. The list of the agricultural commodity futures used in this paper is also not exhaustive. Thus, future research may then explore alternative methodological approaches, use an updated sample period covering the COVID-19 crisis, consider additional commodities (e.g., lumber, sugar, orange juice), and examine implications of the results for devising portfolio strategies.

Credit author statement

- The first and second authors: conceptualization; Data curation; Methodology; Formal analysis; Writing - original draft.
- The third author: Conceptualization; Project administration; Funding acquisition; Supervision; Resources; Software.
- The fourth author: Validation; Writing - review & editing.
- The fifth author: Investigation; Writing - review & editing.

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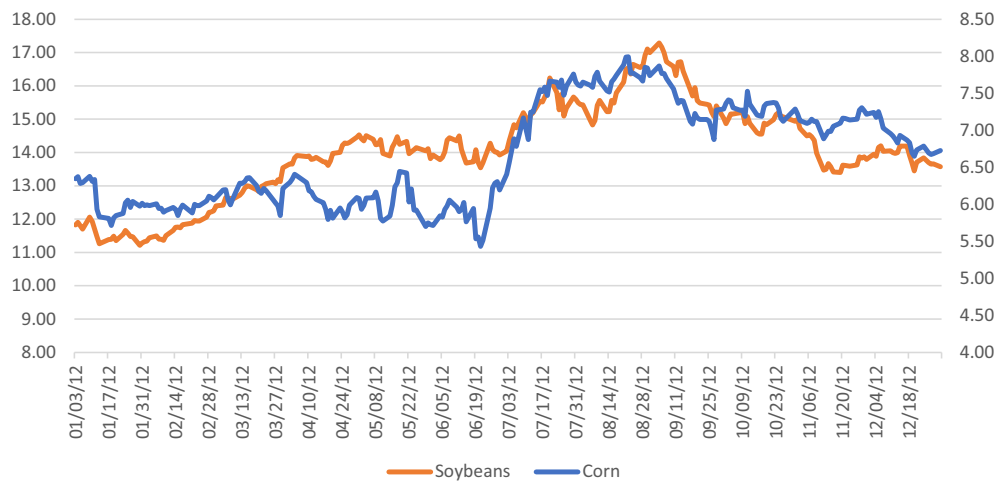


Fig. A1. Corn and soybean futures prices in 2012.

Note. This graph plots daily soybean and corn futures prices in the left and right axis, respectively. The price details are as follows: Exchange: Chicago Board of Trade; Contract unit: 5,000 bushels; Currency: US Dollar; Tick size: \$0.0025 per bushel, \$12.5 per contract; Contract months: March, May, July, September, December; Settlement: Physical.

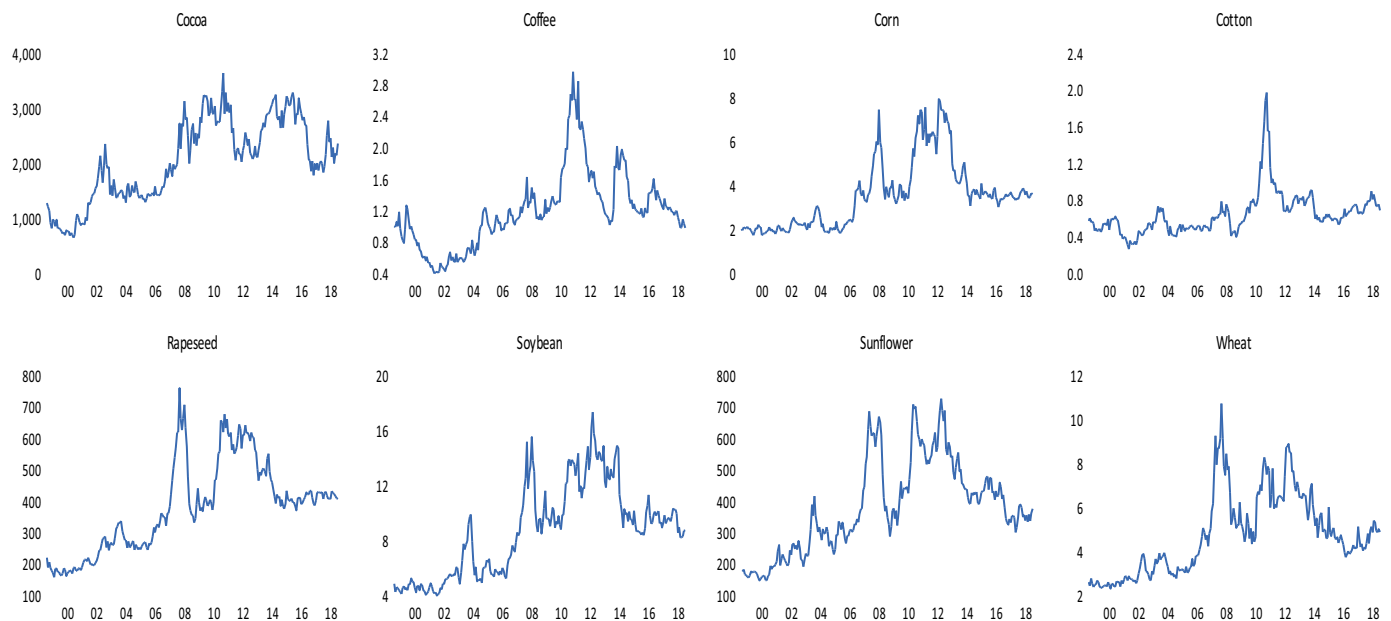


Fig. A2. Time-trends of the agricultural commodity futures prices.

Note. This graph plots monthly agricultural commodity futures prices. The details of the commodity futures are presented in Table 1.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2021.105377>.

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